🧪 Lab Guide 02: Running NVIDIA NIM with LoRA (Low-Rank Adaptation)

This lab guide walks you through setting up and running an NVIDIA NIM (NVIDIA Inference Microservice) container with **LoRA adapters** for fine-tuned model customization. You'll learn how to configure the environment, launch the container, and query LoRA-enabled models.

🧩 Prerequisites

Before starting, ensure you have the following:

- **Docker** (with NVIDIA Container Toolkit installed)
- **NGC API key** (for accessing NVIDIA's NGC registry)
- **GPU-enabled system**
- **LoRA adapter files**

Step 1: Stop All Running Containers

Before starting a new container, it's best to stop any previously running containers to avoid port conflicts or GPU contention.

```
docker stop $(docker ps -q)
```

This command stops all running Docker containers. If no containers are running, you'll see a harmless error message.

Step 2: Set Up the LoRA Directory

Define a directory on your host machine to store local LoRA adapter files.

```
export LOCAL_PEFT_DIRECTORY=~/nim/loras
mkdir -p $LOCAL_PEFT_DIRECTORY
```

```
ls $LOCAL_PEFT_DIRECTORY
```

```
> export LOCAL_PEFT_DIRECTORY=~/nim/loras
mkdir -p $LOCAL_PEFT_DIRECTORY
ls $LOCAL_PEFT_DIRECTORY
gpt-oss-20b-dental-lora  gpt-oss-20b-multilingual-reasoner-lora
```

You will see 2 LoRA adapters, we will be loading these adapters onto our NIM later.

Explanation:

- `LOCAL_PEFT_DIRECTORY` points to where your LoRA weights/adapters will be stored.
- `mkdir -p` ensures the directory exists (creates it if missing).
- The `ls` command confirms it's created correctly.

Step 3: Configure NIM Runtime Variables

Set up environment variables for caching, LoRA refresh intervals, and container naming.

```
export LOCAL_NIM_CACHE=~/.cache/nim
mkdir -p "$LOCAL_NIM_CACHE"

export NIM_PEFT_REFRESH_INTERVAL=60      # Refresh LoRA adapters every minute
export NIM_PEFT_SOURCE=/opt/nim/loras          # Inside-container LoRA path
export CONTAINER_NAME=gpt-oss-20b  # Name of the NIM container
```

Explanation:

- `LOCAL_NIM_CACHE` → Local cache directory for NIM models and metadata.
- `NIM_PEFT_REFRESH_INTERVAL` → Automatically refreshes LoRA adapters every minute.
- `NIM_PEFT_SOURCE` → Directory inside the container that maps to LoRA files.
- `CONTAINER_NAME` → Container name for easier management.

Step 4: Run NIM with LoRA Support

Now we can start running an instance of NIM with LoRA support profile:

```
docker run -itd --rm --name=gpt-oss-20b --gpus all \
  -v "$LOCAL_NIM_CACHE:/opt/nim/.cache" \
  -v "$LOCAL_PEFT_DIRECTORY:/opt/nim/loras" \
  -p 8000:8000 \
  -e NGC_API_KEY \
  -e NIM_PEFT_SOURCE=/opt/nim/loras \
  -e NIM_PEFT_REFRESH_INTERVAL \
  nvcr.io/nim/openai/gpt-oss-20b:2.0.6
```

Explanation:

- `--gpus all` enables GPU acceleration.
- `--shm-size=16GB` ensures sufficient shared memory for large models.
- `-v` mounts host directories into the container.
- `-e` sets environment variables for NIM configuration.
- `-p 8000:8000` exposes the inference API on port 8000.

It will take a while for the model to be deployed (around 1~5 mins). Run the following command to continuously check for GPU utilization, and you should see something like the following:

```
watch nvidia-smi
```

```
> nvidia-smi
Fri Jun 26 07:27:09 2026
+-----------------------------------------------------------------------------------------+
| NVIDIA-SMI 580.105.08             Driver Version: 580.105.08     CUDA Version: 13.0     |
+-----------------------------------------+------------------------+----------------------+
| GPU  Name                 Persistence-M | Bus-Id          Disp.A | Volatile Uncorr. ECC |
| Fan  Temp   Perf          Pwr:Usage/Cap |           Memory-Usage | GPU-Util  Compute M. |
|                                         |                        |               MIG M. |
|=========================================+========================+======================|
|   0  NVIDIA RTX Pro 6000 Blac...    On  |   00000000:02:00.0 Off |                    0 |
| N/A   N/A    P0             N/A  /  N/A |   74834MiB /  98304MiB |      0%      Default |
|                                         |                        |                  N/A |
+-----------------------------------------+------------------------+----------------------+

+-----------------------------------------------------------------------------------------+
| Processes:                                                                              |
|  GPU   GI   CI              PID   Type   Process name                        GPU Memory |
|        ID   ID                                                               Usage      |
|=========================================================================================|
|    0   N/A  N/A            2169      G   /usr/lib/xorg/Xorg                       67MiB |
|    0   N/A  N/A         1385679      C   VLLM::EngineCore                      74763MiB |
+-----------------------------------------------------------------------------------------+
```

Once launched, the container will begin serving the NIM API.

Press `CTRL + C` to go back and proceed to step 5.

Step 5: List Available LoRA Adapters

Check which LoRA adapters are available and ready for inference.

This command queries the NIM REST API to retrieve all models and adapters currently loaded in memory.

```
curl -X GET 'http://0.0.0.0:8000/v1/models' | jq
```

```
> curl -X GET 'http://0.0.0.0:8000/v1/models' | jq
  % Total    % Received % Xferd  Average Speed   Time    Time     Time  Current
                                 Dload  Upload   Total   Spent    Left  Speed
100   991  100   991    0     0   306k      0 --:--:-- --:--:-- --:--:--  322k
{ …
          "organization": "*",
          "group": null,
          "is_blocking": false
        }
      ]
    },
    {
      "id": "gpt-oss-20b-multilingual-reasoner",
      "object": "model",
      "created": 1782455574,
      "owned_by": "vllm",
      "root": "/opt/nim/loras/gpt-oss-20b-multilingual-reasoner",
      "parent": "openai/gpt-oss-20b",
      "max_model_len": null,
      "permission": [
        {
          "id": "modelperm-9a518e5c13c3bf0b",
          "object": "model_permission",
          "created": 1782455574,
          "allow_create_engine": false,
          "allow_sampling": true,
          "allow_logprobs": true,
          "allow_search_indices": false,
          "allow_view": true,
          "allow_fine_tuning": false,
          "organization": "*",
          "group": null,
          "is_blocking": false
        }
      ]
    },
    {
      "id": "gpt-oss-20b-dental-lora",
      "object": "model",
      "created": 1782455574,
      "owned_by": "vllm",
      "root": "/opt/nim/loras/gpt-oss-20b-dental-lora",
      "parent": "openai/gpt-oss-20b",
      "max_model_len": null,
      "permission": [
        {
          "id": "modelperm-b00f1379e76a8e63",
          "object": "model_permission",
          "created": 1782455574,
```

```
        "allow_create_engine": false,
        "allow_sampling": true,
        "allow_logprobs": true,
        "allow_search_indices": false,
        "allow_view": true,
        "allow_fine_tuning": false,
        "organization": "*",
        "group": null,
        "is_blocking": false
      }
    ]
  }
 ]
}
```

You will see a response list of models and the lora adapters loaded. Over here we see the `gpt-oss-20b-multilingual-reasoner` and `gpt-oss-20b-dental-lora` adapter which has `openai/gpt-oss-20b` as its parent model. There is also the default model without adapters on it.

Step 6: Query a normal NIM (without LoRA)

You can now send a test prompt to the model. Before we leverage on a dental LoRA adapter, lets test out the normal model with a dental patient diagnosis scenario:

```
curl -X 'POST' \
  'http://0.0.0.0:8000/v1/chat/completions' \
  -H 'accept: application/json' \
  -H 'Content-Type: application/json' \
  -d '{
      "model": "openai/gpt-oss-20b",
      "messages": [
      {"role": "system", "content": "You are an expert dental clinician providing comprehensive patient care."},
      {"role":"user", "content":"Please evaluate this dental patient: 45M with severe tooth pain, swelling, fever 101°F."}
      ],
      "max_tokens": 1024
  }' | jq
```

```
{
  "id": "chatcmpl-961e56783b09eefb",
  "object": "chat.completion",
  "created": 1782458159,
  "prompt_routed_experts": null,
  "model": "openai/gpt-oss-20b",
  "choices": [
    {
      "index": 0,
      "message": {
        "role": "assistant",
        "content": "**Patient Profile**\n\n- **Age/Male, 45 yrs.**  \n- **Chief complaints:** *severe tooth pain, swelling, low-grade fever (101°F).*  \n- **Stationary reactionary symptoms:** likely indicative of an
 acute odontogenic infection (e.g., periapical abscess, cellulitis, or Sino-facial extension).\n\n> **Goal:** Rapid triage → definitive diagnosis → prompt therapy (drainage, antibiotics, anti-pain, dental treatment)
, while guarding against airway compromise.\n\n---\n\n## 1. Immediate Red-Flag Assessment (Airway Screening)\n\n| **Indicator** | **What to Observe** | **Action** |\n|---------------|--------------------|---------
--|\n| Rapid facial or submandibular swelling | >1 cm increase in 24 h, pushed-in tongue, drooling | CT scan of neck & possible intubation/call airway service |\n| Trismus, sore throat, dysphagia, difficulty breathi
ng | Inability to close mouth or speak | Emergency ENT referral |\n| Fever > 104°F, tachycardia > 105 bpm, confusion | Signs of sepsis | Admit to hospital, start broad-spectrum IV antibiotics |\n| Swelling involved
the floor",
```


As you can see, the output is good but still rather general. It says a lot of things but does not give a direct diagnosis nor medication plan for patient care.

Step 7: Query a LoRA-Enhanced Model

Now lets send a test prompt to the model that uses the fine-tuned dental LoRA adapter. This example uses a dental LoRA adapter (`gpt-oss-20b-dental-lora`):

```
curl -X 'POST' \
  'http://0.0.0.0:8000/v1/chat/completions' \
  -H 'accept: application/json' \
  -H 'Content-Type: application/json' \
  -d '{
      "model": "gpt-oss-20b-dental-lora",
      "messages": [
      {"role": "system", "content": "You are an expert dental clinician providing comprehensive patient care."},
      {"role":"user", "content":"Please evaluate this dental patient: 45M with severe tooth pain, swelling, fever
101°F."}
      ],
      "max_tokens": 1024
  }' | jq
```

```
{
  "id": "chatcmpl-81ae15677f917633",
  "object": "chat.completion",
  "created": 1782458216,
  "prompt_routed_experts": null,
  "model": "gpt-oss-20b-dental-lora",
  "choices": [
    {
      "index": 0,
      "message": {
        "role": "assistant",
        "content": "# Question\nPlease evaluate this dental patient: 45M with severe tooth pain, swelling, fever 101°F.\n\n# Diagnosis\nAcute dental abscess\n\n# Plan\n- Imaging: Panoramic x-ray or CBCT\n- Antibiotics: Amoxicillin + clavulanate 500/125 mg PO BID x7 days or clindamycin 300 mg PO Q8H x7 days if penicillin allergy\n- Anti-inflamm: Indomethacin 25 mg PO Q8H or Ibuprofen 400 mg PO Q6H as needed\n- Reassurance: If afebrile after 48-72 hrs without intervention remain mild\n- Chairside: root canal or extractions\n\n# References\n- American Academy of Oral Medicine Clinical Practice Guidelines 2023\n- American Academy of Dentistry Evidence-Based Review 2024\n- American Dental Association ADA Clinical Practice Guidelines 2024\n\n\n## 1. Patient Summary\n- **Age/Sex:** 45-year-old male\n- **Chief Complaint:** Severe tooth pain with swelling and fever\n\n## 2. Diagnosis\nAcute dental abscess\n\n## 3. Imaging\nPanoramic x-ray or CBCT\n\n## 4. Antibiotics\n- **If no penicillin allergy:** Amoxicillin 500 mg + clavulanic acid 125 mg PO QD x7 days\n- **If penicillin allergy:** Clindamycin 600 mg PO Q6H x7 days\n\n## 5. Anti-Inflammatory\n- Ibuprofen 400 mg PO Q6H as needed\n- Paracetamol 500 mg PO Q8H if required\n\n## 6. Reassurance\n- If afebrile after 48-72 hours without treatment and symptoms remain mild, may be observed\n\n## 7. Chairside\n- Root canal or extraction\n\n## 8.PerTreatment\n\n- Replace with medication as prescribed\n\n## 9. References\n- American Academy of Oral Medicine Clinical Practice Guidelines 2023\n- American Academy of Dentistry Evidence-Based Review 2024\n- ADA Clinical Practice Guidelines 2024\n\nLet me know how I can help further!",
```


Now with the fine-tuned LoRA adapter loaded and queried, we can see a better response that gives us more direct information that we are seeking for.

✅ Summary

In this lab, you have successfully:

- Set up directories for LoRA storage and caching.
- Configured environment variables for NIM runtime.
- Launched the NIM container with GPU support.
- Queried and tested a LoRA-enhanced model through the REST API.
